

TABLE

ABLATION RESULTS FOR DIFFERENT MODULE COMBINATIONS OF RMT-D. MODULE A DENOTES A T-W-REFERENCED SEQUENTIAL PROCEDURE FOR JOINT EFFECTIVE-RANK AND NOISE-SCALE ESTIMATION. MODULE B DENOTES VOXEL-WISE SPATIAL-SUPPORT SELECTION. MODULE C DENOTES ALIGNMENT-AWARE SINGULAR-VALUE SHRINKAGE. WHEN THE CORRESPONDING MODULES ARE DISABLED, RANK AND NOISE ESTIMATION BASED ON THE M-P SPECTRAL EDGE, A FIXED TWO-DIMENSIONAL 5×5 SPATIAL SUPPORT, AND CONVENTIONAL SINGULAR-VALUE SHRINKAGE WITHOUT ALIGNMENT AWARENESS ARE USED, RESPECTIVELY. A+B+C DENOTES THE COMPLETE RMT-D. RESULTS ARE REPORTED AS THE MEAN ± STANDARD DEVIATION ACROSS 30 INDEPENDENT REALIZATIONS. RED AND BLUE MEAN VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY, WITHIN EACH NOISE CONDITION. RANKINGS FOR $\Delta tSNR$ ARE BASED ON THE ABSOLUTE DISTANCE FROM 0.

(a) Gaussian

Noise	Module	MSE↓	PSNR↑	$\Delta tSNR \rightarrow 0$	FC_MAE↓	FC_F↓	r↑
Gaussian=2	A	0.2681 ± 0.0011	67.586 ± 0.0304	5.5269 ± 1.4162	0.0126 ± 0.0003	1.6232 ± 0.0475	0.9773 ± 0.0010
	B	0.1761 ± 0.0005	69.410 ± 0.0255	1.7821 ± 0.2071	0.0120 ± 0.0003	1.6412 ± 0.0485	0.9847 ± 0.0007
	C	0.2680 ± 0.0011	67.587 ± 0.0303	5.5590 ± 1.4122	0.0127 ± 0.0003	1.6401 ± 0.0478	0.9773 ± 0.0010
	A+B	0.1766 ± 0.0005	69.398 ± 0.0255	1.5943 ± 0.1477	0.0116 ± 0.0002	1.5783 ± 0.0474	0.9846 ± 0.0007
	A+C	0.2678 ± 0.0011	67.590 ± 0.0302	5.5391 ± 0.3314	0.0129 ± 0.0003	1.6573 ± 0.0484	0.9774 ± 0.0010
	B+C	0.1772 ± 0.0005	69.383 ± 0.0257	1.5989 ± 0.1362	0.0115 ± 0.0003	1.5724 ± 0.0498	0.9846 ± 0.0007
	A+B+C	0.1252 ± 0.0004	70.956 ± 0.0244	0.9345 ± 0.2817	0.0027 ± 0.0004	0.4829 ± 0.0510	0.9884 ± 0.0005
Gaussian=5	A	1.2335 ± 0.0126	60.958 ± 0.0475	26.919 ± 2.5011	0.0543 ± 0.0013	6.8219 ± 0.1769	0.9124 ± 0.0035
	B	0.8025 ± 0.0056	62.825 ± 0.0337	10.255 ± 0.7968	0.0624 ± 0.0011	8.4934 ± 0.1882	0.9429 ± 0.0022
	C	1.2337 ± 0.0125	60.957 ± 0.0473	27.072 ± 2.4908	0.0549 ± 0.0013	6.8985 ± 0.1811	0.9124 ± 0.0035
	A+B	0.8055 ± 0.0055	62.808 ± 0.0334	9.6480 ± 0.7600	0.0604 ± 0.0011	8.2164 ± 0.1907	0.9424 ± 0.0022
	A+C	1.2346 ± 0.0126	60.954 ± 0.0476	28.706 ± 3.3191	0.0561 ± 0.0013	7.0574 ± 0.1788	0.9124 ± 0.0035
	B+C	0.8088 ± 0.0058	62.791 ± 0.0350	9.4654 ± 0.7176	0.0597 ± 0.0010	8.0928 ± 0.1875	0.9421 ± 0.0022
	A+B+C	0.6157 ± 0.0053	63.987 ± 0.0338	1.9506 ± 0.4682	0.0309 ± 0.0013	4.8617 ± 0.1946	0.9557 ± 0.0017
Gaussian=10	A	3.5443 ± 0.0550	56.374 ± 0.0666	201.54 ± 121.07	0.1500 ± 0.0075	24.905 ± 1.3531	0.8029 ± 0.0066
	B	2.3850 ± 0.0335	58.095 ± 0.0603	49.834 ± 17.090	0.1545 ± 0.0044	23.457 ± 0.7411	0.8752 ± 0.0042
	C	3.5458 ± 0.0548	56.373 ± 0.0662	200.39 ± 120.27	0.1525 ± 0.0075	25.211 ± 1.3531	0.8025 ± 0.0066
	A+B	2.3842 ± 0.0332	58.096 ± 0.0603	36.232 ± 2.7774	0.1511 ± 0.0042	22.667 ± 0.7222	0.8762 ± 0.0042
	A+C	3.5503 ± 0.0555	56.367 ± 0.0671	204.99 ± 95.581	0.1574 ± 0.0077	26.338 ± 1.3718	0.8056 ± 0.0066
	B+C	2.3919 ± 0.0335	58.082 ± 0.0604	35.992 ± 2.7237	0.1498 ± 0.0041	21.428 ± 0.6997	0.8754 ± 0.0042
	A+B+C	2.1467 ± 0.0299	58.513 ± 0.0601	3.2238 ± 0.6414	0.1497 ± 0.0050	21.337 ± 0.6832	0.8923 ± 0.0036

(b) Rician

Noise	Module	MSE↓	PSNR↑	Δ tSNR→0	FC_MAE↓	FC_F↓	r↑
Rician=2	A	0.2684 ± 0.0010	67.580 ± 0.0297	4.9744 ± 0.4389	0.0125 ± 0.0003	1.6095 ± 0.0502	0.9774 ± 0.0010
	B	0.1763 ± 0.0004	69.406 ± 0.0268	1.7623 ± 0.1705	0.0120 ± 0.0003	1.6280 ± 0.0727	0.9847 ± 0.0006
	C	0.2686 ± 0.0010	67.578 ± 0.0302	5.0061 ± 0.4331	0.0127 ± 0.0003	1.6264 ± 0.0501	0.9773 ± 0.0010
	A+B	0.1768 ± 0.0004	69.394 ± 0.0271	1.5965 ± 0.1267	0.0115 ± 0.0002	1.5657 ± 0.0715	0.9846 ± 0.0006
	A+C	0.2682 ± 0.0009	67.584 ± 0.0296	5.1686 ± 0.5350	0.0128 ± 0.0003	1.6436 ± 0.0509	0.9775 ± 0.0010
	B+C	0.1774 ± 0.0004	69.379 ± 0.0266	1.6180 ± 0.1360	0.0115 ± 0.0003	1.5600 ± 0.0712	0.9846 ± 0.0006
	A+B+C	0.1254 ± 0.0003	70.956 ± 0.0265	0.9182 ± 0.2486	0.0028 ± 0.0005	0.4830 ± 0.0784	0.9883 ± 0.0005
Rician=5	A	1.2371 ± 0.0120	60.945 ± 0.0469	27.200 ± 2.1247	0.0543 ± 0.0012	6.7970 ± 0.1802	0.9121 ± 0.0034
	B	0.8048 ± 0.0051	62.812 ± 0.0359	10.419 ± 0.5458	0.0621 ± 0.0013	8.3818 ± 0.2606	0.9427 ± 0.0022
	C	1.2392 ± 0.0120	60.938 ± 0.0467	27.478 ± 1.8915	0.0548 ± 0.0012	6.8708 ± 0.1887	0.9119 ± 0.0034
	A+B	0.8081 ± 0.0050	62.795 ± 0.0356	9.8130 ± 0.4951	0.0602 ± 0.0013	8.1081 ± 0.2557	0.9423 ± 0.0023
	A+C	1.2382 ± 0.0119	60.942 ± 0.0465	25.150 ± 4.3641	0.0560 ± 0.0011	7.0341 ± 0.1836	0.9121 ± 0.0034
	B+C	0.8111 ± 0.0054	62.778 ± 0.0362	9.5976 ± 0.4848	0.0594 ± 0.0013	7.9815 ± 0.2575	0.9420 ± 0.0023
	A+B+C	0.6286 ± 0.0046	63.935 ± 0.0363	1.9661 ± 0.4648	0.0226 ± 0.0015	4.7426 ± 0.2676	0.9554 ± 0.0019
Rician=10	A	3.5736 ± 0.0539	56.339 ± 0.0666	202.63 ± 105.89	0.1510 ± 0.0038	24.861 ± 0.7439	0.8016 ± 0.0068
	B	2.4119 ± 0.0326	58.046 ± 0.0624	41.871 ± 6.4136	0.1544 ± 0.0035	23.445 ± 0.7433	0.8746 ± 0.0043
	C	3.5933 ± 0.0515	56.315 ± 0.0631	168.89 ± 27.931	0.1506 ± 0.0038	25.147 ± 0.7345	0.8008 ± 0.0068
	A+B	2.4148 ± 0.0333	58.041 ± 0.0638	36.597 ± 4.0591	0.1498 ± 0.0050	22.587 ± 0.7041	0.8754 ± 0.0043
	A+C	3.5820 ± 0.0545	56.328 ± 0.0671	226.63 ± 103.05	0.1576 ± 0.0038	26.315 ± 0.7269	0.8042 ± 0.0068
	B+C	2.4221 ± 0.0333	58.028 ± 0.0631	36.567 ± 5.5408	0.1584 ± 0.0051	22.405 ± 0.7159	0.8745 ± 0.0043
	A+B+C	2.1796 ± 0.0283	58.486 ± 0.0597	3.8532 ± 0.6922	0.1491 ± 0.0020	21.224 ± 0.2681	0.8916 ± 0.0038